

DATA CLEANING & PREPROCESSING REPORT

Social Media Sentiment Analysis Internship Track — Week 2 Deliverable

Project Name:

Social Media Sentiment Analysis

Milestone:

Task 1 to Task 3 (Data Collection, Cleaning, & Storage)

Date:

June 18, 2026

1. Project Objective & Architecture

The core goal of this phase was to construct a robust automated preprocessing pipeline to convert noisy, unformatted social media inputs into reliable structured data fit for training subsequent sentiment classification models. The workspace architecture follows structural documentation requirements precisely:

- `data/` — Contains raw tracking datasets and the newly prepared output records.
- `src/` — Houses operational modular Python application scripts (`main.py`).
- `reports/` — Dedicated hub for historical tracking documentation and analytical summaries.

2. Executed Data Engineering Interventions

Using the Python pandas library, a linear operational pipeline was implemented to clean inconsistencies out of the dataset:

- Missing Value Resolution:** Evaluated rows against essential metadata tracking. Automatically isolated and purged records that lacked required structural conversational data (such as the completely blank text element on row 104).
- Redundancy Filtering:** Detected identical multi-post scrapes and retweets. Dropped duplicate text fields to ensure that every sample represents a unique text data point.
- Text Standardization:** Stripped external white spaces and mapped strings uniformly to lowercase to prepare tokens for vocabulary indexing and feature engineering.
- Data Format Fixes:** Manually troubleshooted comma spacing errors on unquoted data strings (Row 105) that broke structural token parsing, ensuring full file read compatibility.

3. Before vs. After Cleaning Metrics

Data Metric	Raw Source Status	Processed Target Status	Pipeline Action Taken
Total Rows File Size	5 rows	3 rows	-2 invalid records filtered out

Data Metric	Raw Source Status	Processed Target Status	Pipeline Action Taken
Blank Text Fields	1 missing record (104)	0 null elements	Purged via automated logic
Duplicate Profiles	1 duplicate record (103)	0 duplicate entries	Deduplicated systematically
Text Formats	Mixed casing structure	Standardized lowercase	Mapped uniformly via string methods

4. Conclusion & Storage Status

The final target database has been successfully formatted and saved to the storage directory as `cleaned_social_media_data.csv`. With all blank entries, duplication artifacts, and syntax tokenization errors fully resolved, the data layer satisfies all foundational performance benchmarks and is verified ready for advanced training vectorizers.